

Product Requirements Document (PRD)

Project Name: Clever Cubes

1. Project Overview

The objective of this project is to build a responsive, web-based platform named Clever Cubes that offers a variety of interactive puzzle games. The website will allow users to play puzzles for free, track their completion times, and compete on leaderboards. The platform must be fast, user-friendly, and accessible on both desktop and mobile web browsers.

2. Target Audience

- **Casual Gamers:** People looking for a quick mental break or entertainment.
- **Puzzle Enthusiasts:** Users who actively seek out brain-teasers, logic puzzles, and daily challenges.
- **All Age Groups:** The UI should be intuitive enough for children and accessible enough for older adults.

3. Goals and Success Metrics

- **Goal 1:** Launch a fully functional MVP (Minimum Viable Product) featuring at least 3 distinct puzzle types.
- **Goal 2:** Ensure seamless responsive gameplay across desktop, tablet, and mobile devices.
- **Goal 3:** Establish a user progression system (saving scores and times).
- **Success Metrics:** Smooth gameplay with zero lag, successful user registrations, and fast page load times (under 2 seconds).

4. Scope of Work (MVP Features)

The initial scope will focus on the core functionality necessary to browse, play, and track puzzle progress on Clever Cubes.

4.1. Puzzle Types Included in MVP

1. **Sudoku:** Standard 9x9 grid with dynamic difficulty generation (Easy, Medium, Hard). Includes functionality for note-taking (pencil marks), hint system, and mistake highlighting.

2. **Word Search:** Dynamically generated word search grids based on specific themes (e.g., Animals, Science, Geography). Must support horizontal, vertical, and diagonal word placement.
3. **Sliding Tile Puzzle:** Classic 15-puzzle (4x4 grid) or 8-puzzle (3x3 grid) where users slide numbered tiles into numerical order.

4.2. User Roles

- **Guest User:** Can play puzzles but cannot save progress, appear on leaderboards, or track history.
- **Registered User:** Can play puzzles, save progress, earn scores, appear on leaderboards, and view play history.
- **Administrator:** Has access to a backend dashboard to manage users, add new puzzle data (e.g., new word lists for Word Search), and monitor site health.

5. Functional Requirements

5.1. Front-End / User Interface

- **Homepage:**
 - A clean, welcoming landing page featuring "Daily Challenges" and a categorized list of available puzzle types.
 - Clear Call-to-Action (CTA) for users to register or start playing immediately.
- **Puzzle Category Pages:**
 - Displays a list of available puzzles within a specific category, filterable by difficulty.
- **Game Interface:**
 - **Play Area:** The central interactive grid for the specific puzzle.
 - **Timer:** A visible clock tracking how long the user takes to complete the puzzle.
 - **Controls:** Buttons for "Pause", "Restart", "Hint" (if applicable), and "Submit/Check".
 - **Completion Screen:** A pop-up or screen transition showing the final time, score, and a "Play Again" or "Next Puzzle" button.
- **User Authentication System:**
 - Sign up / Log in using Email/Password.
 - Password reset functionality ("Forgot Password").
- **User Profile / Dashboard:**
 - Displays basic user info (Username, Join Date).
 - Shows personal statistics: Total puzzles solved, best times for each difficulty, and recent play history.
- **Global Leaderboards:**
 - Rankings based on fastest completion times for specific puzzles and difficulties. Filterable by "Today", "This Week", and "All Time".

5.2. Back-End / Admin Interface

- **User Management:** View registered users, reset passwords, ban/delete accounts if necessary.
- **Content Management:**
 - Add/Edit/Delete themes and word lists for the Word Search puzzle.
 - Adjust point values or parameters for the Sudoku generator.
- **Database:**
 - Store user credentials securely (hashed passwords).
 - Store puzzle completion logs (User ID, Puzzle Type, Difficulty, Time Taken, Timestamp).

6. Non-Functional Requirements

6.1. Performance & Responsiveness

- **Mobile-First Design:** The puzzle grids must scale perfectly to fit mobile screens without horizontal scrolling. Touch interactions (tapping, swiping) must be highly responsive.
- **Speed:** Game interfaces must load rapidly. Puzzle logic (generation of Sudoku grids or Word Searches) should happen almost instantaneously.

6.2. Security

- All data transmission must be encrypted via HTTPS/SSL.
- User passwords must be securely hashed (e.g., using bcrypt).
- Protection against basic web vulnerabilities (SQL Injection, XSS, CSRF).
- Anti-cheat mechanisms: Basic validation on the server side to ensure submitted completion times are humanly possible.

6.3. Accessibility

- High contrast color themes for readability.
- Clear typography.
- Keyboard navigability for puzzles where applicable (e.g., using arrow keys and numbers for Sudoku).

7. Recommended Technology Stack

(Note to Freelancer: You may propose alternatives, but this is the preferred baseline)

- **Front-End:** React.js or Vue.js (ideal for managing the complex state of puzzle grids), HTML5, CSS3/Tailwind CSS.
- **Back-End:** Node.js with Express, or Python (Django/FastAPI).

- **Database:** PostgreSQL or MongoDB.
- **Hosting:** AWS, Vercel, or Heroku.

8. Deliverables

1. **Source Code:** Fully commented and structured code pushed to a provided GitHub repository.
2. **Database Schema:** Documentation of the database structure.
3. **Deployment:** Setup of the application on the chosen production server.
4. **Admin Documentation:** A brief text guide explaining how to access the admin panel and manage puzzle data.